

Fig. 5: Actual-size PCB pattern for the input power supply

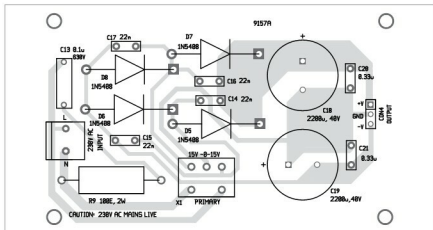


Fig. 6: Component layout for the PCB in Fig. 5

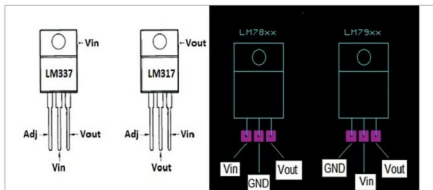


Fig. 7: Pin details of voltage regulators

separate input power supply section from the regulator section. Then you can change one part of the power supply and keep the other intact. It helps in making different combinations with the rectifying and regulator sections.

Construction and testing

An actual size, single-side PCB pattern for universal power supply with LM317 and LM337 is shown in Fig. 3 and its component layout in Fig. 4. Actual-size, single-side PCB pattern for the input power supply section is

shown in Fig. 5 and its component layout in Fig. 6.

After assembling the circuit, connect the supply from CON4 to CON1. Now positive and negative outputs can be adjusted separately. Pin details of voltage regulators used in Fig. 1 are shown in Fig. 7.

ZD1 is for LM317, ZD2 is for LM337. Jumper 2 (J2) controls the applied voltage for the ZD1 and LM317. Jumper 1 (J1) controls the applied voltage for the lower Zener ZD2 and LM337.

When J1 is open and 1N4731 Zener is used (no shorting). ZD2 for LM337 operates as simple diode with voltage drop of around 0.7V. The forward current is only from VR2 and R6.

When J1 is shorted between points 1 and 2. ZD2 for LM337 operates as a simple Zener diode with voltage drop of around 4.3V. The current is from R3 and VR2.

When J1 is shorted between points 2 and 3. ZD2 for LM337 operates as a simple diode with voltage drop around 0.7V, but with higher forward current (compared to the case of no shorting).

Similarly, you can get various outputs by shorting J2 between points 1, 2 and 3.

Glowing of LED3 and LED4 indicates the output status of LM317 and LM337, respectively.

EFY note. 1. 79xx can replace LM337, and 78xx can replace LM317, in the same PCB with additional wiring. Please refer datasheets of both the ICs before making changes.

2. We have not tested using LM78xx and LM79xx regulators, so the PCB shown here is designed for LM317 and LM337 only. **EFY**

